

fn get_margin

Michael Shoemate (@Shoeboxam)

This proof resides in “contrib” because it has not completed the vetting process.

Proves soundness of `get_margin` in `mod.rs` at commit `f5bb719` (outdated¹).

`get_margin` returns a `Margin` for a given set of grouping columns (`by`) whose descriptors are no more restrictive than what is known in `FrameDomain`.

1 Hoare Triple

Precondition

Compiler-verified

Types matching pseudocode.

Caller-verified

None

Pseudocode

```
1 from math import prod
2
3 def get_margin(domain: FrameDomain, by: set[Expr]) -> Margin:
4     margin = next( #
5         (m for m in domain.margins if m.by == by),
6         Margin.by(by)
7     )
8
9     subset_margins = [ #
10         margin
11         for margin in domain.margins
12         if margin.by.issubset(by)
13     ]
14
15     margin.max_length = min( #
16         m.max_length
17         for m in subset_margins
18         if m.max_length is not None
19     )
20
21     all_max_groups = [ #
22         (m.by, m.max_groups)
23         for m in domain.margins
24         if m.max_groups is not None
25     ]
```

¹See new changes with `git diff f5bb719..5109caa rust/src/domains/polars/frame/mod.rs`

```

26 max_groups_covering = find_min_covering(grouping_columns, all_max_groups)
27 if max_groups_covering is not None:
28     margin.max_groups = prod(v for _, v in max_groups_covering)
29
30 all_invariants = ( #
31     m.invariant
32     for m in domain.margins
33     if by.issubset(m.by)
34 )
35 margin.invariant = max(all_invariants, key={None: 0, "Keys": 1, "Lengths": 2}.get)
36
37 return margin

```

Postcondition

Theorem 1.1. Returns a `Margin` that describes properties of members of `domain` when grouped by `by`.

Proof. On line 4, `margin` is either a valid margin descriptor for `by`, by the definition of `domain`, or it is the default margin, which is a valid margin descriptor for all potential datasets.

We now update descriptors based on other information available in `domain`.

On line 9, `subset_margins` is the subset of margins spanned by `by`. Then 15 assigns the smallest known descriptors over any margin spanning a subset of the grouping columns.

If `max_length` is known about a coarser data grouping (when grouped by fewer columns), then these descriptors still apply to a finer data grouping, as group length can only decrease when more finely splitting data. Therefore `max_length` remains valid after mutation.

Line 21 retrieves all known `max_groups` descriptors. There are no manual preconditions to `find_min_covering`, therefore we claim the postcondition, that the output is a covering for `by`.

The number of groups can be no greater than the cardinality of the cartesian product of the keys of each grouping column. Therefore the code finds a set of `max_groups` descriptors that covers the grouping columns, and then updates `margin` to the product of the covering.

On an aside, for utility, while the covering found may not be the smallest, the greedy algorithm will always choose a singleton cover if it is available, therefore this update to the descriptor cannot increase the descriptor.

Finally, on line 30, `all_invariants` contains invariants for any margin finer than `by`. If group keys and/or lengths are known for a finer grouping, then they are also valid for a coarser grouping. Therefore `invariants` is updated to the strongest invariant for any grouping as fine or finer than `by`.

Since the initial margin (4) was valid, and all updates have also been shown to be valid, `get_margin` returns a `Margin` that describes properties of members of `domain` when grouped by `by`. $\square$